Here is the clinical evidence extracted from the **Global Strategy for Asthma Management and Prevention (2026 update)** for use in a medical RAG system.

Topic

ASTHMA DIAGNOSIS

Clinical Question

How is asthma diagnosed and what are the objective criteria for confirming variable airflow limitation?

Recommendation

The diagnosis of asthma is based on the identification of both a characteristic pattern of respiratory symptoms and the presence of variable expiratory airflow limitation 1. Objective tests must be used to confirm the diagnosis, preferably before treatment is started 1, 2.

Criteria / Thresholds

1. History of Typical Respiratory Symptoms:

- **Symptoms:** Wheeze, shortness of breath, chest tightness, and cough 2.
- **Supportive Features:** Symptoms vary over time and in intensity; often worse at night or on waking; triggered by exercise, laughter, allergens, or cold air; worsen after stopping exercise; worsen with viral infections 2.

2. Objective Evidence of Variable Expiratory Airflow Limitation (One or more of the following):

- **Positive Bronchodilator (BD) Responsiveness test:**
- **Adults:** Increase in \\(\text{FEV}_1\\) of **\\(\ge 12\%\\) and \\(\ge 200\text{ mL}\\)** from baseline, 10–15 min after 200–400 mcg salbutamol or equivalent 3.
- **Children (6–17 years):** Increase in \\(\text{FEV}_1\\) of **\\(\ge 12\%\\) predicted** 3.
- **Excessive diurnal PEF variability (twice-daily readings over 2 weeks):**
- **Adults:** Average daily diurnal PEF variability **\\(> 10\%\\)** 3.
- **Children:** Average daily diurnal PEF variability **\\(> 13\%\\)** 3.
- **Significant increase in lung function after 4 weeks of controller treatment:**
- **Adults:** Increase from baseline in \\(\text{FEV}_1\\) by **\\(\ge 12\%\\) and \\(\ge 200\text{ mL}\\)** (or PEF by **\\(\ge 20\%\\)**) after 4 weeks of daily ICS-containing treatment 3.
- **Children:** Increase from baseline in \\(\text{FEV}_1\\) of **\\(\ge 12\%\\) predicted** (or in PEF of **\\(\ge 15\%\\)**) 3.
- **Positive Bronchial Challenge Test:**
- **Adults:** Fall from baseline in \\(\text{FEV}_1\\) of **\\(\ge 20\%\\)** with standard doses of methacholine, or **\\(\ge 15\%\\)** with standardized hyperventilation, hypertonic saline, or mannitol challenge, or **\\(> 10\%\\) and \\(> 200\text{ mL}\\)** with standardized exercise challenge 3.

- **Children:** Fall from baseline in FEV_1 of **$\ge 20\%$** with standard doses of methacholine, or **$> 12\%$ predicted** (or fall in PEF **$> 15\%$**) with standardized exercise challenge 3.

3. Supporting Biomarkers (Not diagnostic on their own):

- **Elevated FeNO:** Adults/adolescents: **$> 50\text{ ppb}$**; children: **$> 35\text{ ppb}$** 3, 4.
- **Blood Eosinophils:** Above national/regional reference range 3.

Evidence

"Making the diagnosis of asthma before treatment is started (Box 1-1, p.27 and Box 1-2, p.28) is based on identifying both a characteristic pattern of respiratory symptoms such as wheezing, shortness of breath (dyspnea), chest tightness or cough, and variable expiratory airflow." 1"The greater the variations, or the more occasions excess variation is seen, the more confidently the diagnosis of asthma can be made." 3

Source

- GINA section: Section 1
- Page: 26, 27, 28, 32
- Table/Figure: Box 1-1, Box 1-2

Topic

ASTHMA ASSESSMENT

Clinical Question

What criteria are used to assess asthma symptom control and future risk of adverse outcomes?

Recommendation

Assessment of asthma includes evaluating both symptom control and the risk of adverse outcomes (exacerbations, lung function decline, and medication side-effects) 5.

Criteria / Thresholds

1. Assessment of Recent Symptom Control (Past 4 weeks):

- **Well controlled:** None of the following:
- Daytime asthma symptoms **more than twice/week**.
- Any night waking due to asthma.
- SABA reliever for symptoms **more than twice/week** (excludes pre-exercise).
- Any activity limitation due to asthma 6.
- **Partly controlled: 1–2** of the above criteria 6.
- **Uncontrolled: 3–4** of the above criteria 6.

2. Assessment of Future Risk (Independent of symptom control):

- **Risk factors for exacerbations:** History of **≥ 1 severe exacerbation** in the previous year; SABA-only treatment; over-use of SABA (**≥ 3 canisters/year**); poor adherence; incorrect inhaler technique; low FEV_1 (**$< 60\%$ predicted**); smoking; elevated blood eosinophils or FeNO 6-8.
- **Risk factors for persistent airflow limitation:** Preterm birth; low birth weight; chronic mucus hypersecretion; lack of ICS treatment; tobacco smoke/chemical exposures; low initial FEV_1; blood eosinophilia 9.

3. Lung Function Assessment:

- Measure at diagnosis, **3–6 months** after starting treatment, then periodically (at least every **1–2 years**, more often in high-risk patients) 10, 11.

Evidence

"Asthma control is the extent to which the features of asthma have been reduced or resolved by treatment. It is assessed in two domains: symptom control and risk of adverse outcomes." 5"A low FEV_1 percent predicted: Identifies patients at risk of asthma exacerbations, independent of symptom levels, especially if FEV_1 is $< 60\%$ predicted." 8

Source

- GINA section: Section 2
- Page: 40, 41, 47, 48
- Table/Figure: Box 2-1, Box 2-2

Topic

ASTHMA TREATMENT

Clinical Question

What are the preferred and alternative treatment options for asthma management across the stepwise tracks?

Recommendation

GINA recommends that all adults and adolescents receive ICS-containing treatment to reduce risk of severe exacerbations 12. **Track 1 (Preferred)** uses low-dose ICS-formoterol as the reliever; **Track 2 (Alternative)** uses SABA or ICS-SABA as the reliever 13, 14.

Criteria / Thresholds

1. Adults and Adolescents Stepwise Treatment:

- **Step 1–2:**
- **Track 1 (Preferred):** As-needed only low-dose ICS-formoterol 15, 16.
- **Track 2 (Alternative):** As-needed low-dose ICS taken whenever SABA is taken, or daily low-dose ICS plus as-needed reliever 15.

- **Step 3:**
- **Track 1 (Preferred):** Low-dose maintenance-and-reliever therapy (MART) with ICS-formoterol 15.
- **Track 2 (Alternative):** Low-dose maintenance ICS-LABA plus as-needed reliever 15.
- **Step 4:**
- **Track 1 (Preferred):** Medium-dose MART with ICS-formoterol 15.
- **Track 2 (Alternative):** Medium-dose maintenance ICS-LABA plus as-needed reliever 15.
- **Step 5 (Tracks 1 and 2):** Refer for assessment and phenotyping. Consider add-on LAMA or biologic (anti-IgE, anti-IL5/5R, anti-IL4R, anti-TSLP) 15, 17, 18.

2. Initial Pharmacological Treatment Selection (Adults/Adolescents):

- **Infrequent symptoms (\\(< 2\\) days/week):** Track 1: As-needed low-dose ICS-formoterol 19.
- **Symptoms \\(3–5\\) days/week:** Step 2 (AIR-only Track 1, or daily low-dose ICS Track 2) 20.
- **Symptoms most days, or waking \\(\ge 1\\) night/week, or low lung function:** Step 3 (MART Track 1, or low-dose ICS-LABA Track 2) 21.
- **Daily symptoms, or waking \\(\ge 1\\) night/week, and low lung function or recent exacerbation:** Step 4 (Medium-dose MART Track 1, or medium-dose ICS-LABA Track 2) 22.

3. Stepping Up and Down:

- **Step Up:** If symptoms or exacerbations persist despite good adherence and technique 23.
- **Step Down:** When control is maintained for **2–3 months** 24, 25. Reduce ICS dose by **\\(25–50\%\\)** at **3-month** intervals 26.

Evidence

"Track 1, in which the reliever is low-dose ICS-formoterol, is the preferred approach recommended by GINA. ... This approach is preferred because it reduces the risk of severe exacerbations, compared with using a SABA reliever." 13"Stepping down ICS doses by \\(25–50\%\\) at 3-month intervals is feasible and safe for most patients." 26

Source

- GINA section: Section 4
- Page: 81, 82, 83, 84, 92, 112
- Table/Figure: Box 4-4, Box 4-5, Box 4-6, Box 4-13

Topic

INHALER TECHNIQUE AND ADHERENCE

Clinical Question

How should clinicians assess and improve inhaler technique and adherence?

Recommendation

Clinicians must assess inhaler technique by watching the patient use their device and evaluate adherence using empathic questioning at every visit 27-29.

Criteria / Thresholds

1. Inhaler Technique Management:

- **Watch** the patient use their inhaler and check against a checklist 27.
- **Show** the correct method and have the patient demonstrate back; re-check up to **3 times** if necessary 27.
- **Spacer Use:** Strongly recommended with pMDIs for ICS delivery to reduce local/systemic side-effects 30, 31.
- **Suspension Inhalers:** Salbutamol (albuterol) must be **shaken immediately before each puff** 32, 33.

2. Adherence Assessment:

- **Empathic Questions:** "Many patients don’t use their inhaler as prescribed. In the last 4 weeks, how many days a week have you been taking it?" 27, 34.
- **Verification:** Check dates on inhalers and view dispensing data 34.

3. Causes of Poor Adherence:

- **Unintentional:** Instruction misunderstanding, forgetfulness, lack of routine, cost 35.
- **Intentional:** Perception treatment is unnecessary, concerns about side-effects, denial of diagnosis, cost 35.

Evidence

"Incorrect inhaler technique (seen in up to \\(80\\%\\) patients): Ask the patient to show you how they use their inhaler; compare with a checklist or video." 34"Suboptimal adherence (up to \\(75\\%\\) asthma patients): Ask empathically about frequency of use..." 34

Source

- GINA section: Section 2, Section 5, Section 8
- Page: 40, 52, 119, 121, 156, 158
- Table/Figure: Box 2-4, Box 5-2, Box 5-3

Topic

ASTHMA EXACERBATIONS

Clinical Question

What are the clinical criteria for assessing asthma exacerbation severity and the requirements for initial management and discharge?

Recommendation

Exacerbations are an acute worsening in symptoms and lung function requiring a treatment change 36, 37. Initial management includes SABA, oxygen to target thresholds, and systemic corticosteroids for moderate-to-severe cases 38, 39.

Criteria / Thresholds

1. Severity Assessment (Adults, Adolescents, Children 6+):

- **Mild:** Talks in sentences; can lie down; Pulse **$< 100\text{ bpm}$**; O_2 saturation **$\ge 94\%$**; PEF **$> 70\%$ predicted** 38.
- **Moderate:** Talks in phrases; sits; Resp rate increased but **$\le 30/\text{min}$**; Pulse **$100–120\text{ bpm}$**; O_2 saturation **$\ge 92\%$**; PEF **$50–70\%$ predicted** 38.
- **Severe:** Talks in words; hunched; Resp rate **$> 30/\text{min}$**; Accessory muscles used; Pulse **$> 120\text{ bpm}$**; O_2 saturation **$< 92\%$**; PEF **$< 50\%$ predicted** 38.
- **Life-threatening:** Drowsy, confused; silent chest 38.

2. Management Medications and Doses:

- **SABA (Salbutamol 100 mcg/puff): 4–10 puffs** by pMDI + spacer every **20 min for 1 hour** 38.
- **Systemic Corticosteroids (OCS):**
- **Adults/Adolescents:** Prednisone/prednisolone **$40–50\text{ mg/day}$** for **$5–7\text{ days}$** 40.
- **Children (6–11): $1–2\text{ mg/kg/day}$** (max **40 mg**) for **$3–5\text{ days}$** 40.
- **Oxygen Targets:** Titrate to maintain saturation **$93–95\%$** in adults and **$94–98\%$** in children 38, 41. Supplemental oxygen is not recommended unless saturation is **below 92%** 32.
- **Antibiotics:** Not routinely prescribed for asthma exacerbations 42.

3. Discharge and Follow-up:

- **Discharge Criteria:** Symptoms improved; SABA not needed; O_2 sat **$> 92\%$** on room air; PEF **$> 70\%$** 38.
- **Discharge Medications:** Preferred regimen is Step 4 MART with ICS-formoterol 43, 44.
- **Follow-up:** Within **$2–7\text{ days}$** for adults and **$2–5\text{ days}$** for children 38, 45.

Evidence

"Assess exacerbation severity from the degree of dyspnea, respiratory rate, oxygen saturation and lung function (usually PEF), while starting SABA and oxygen therapy if needed." 46"Supplemental oxygen not recommended unless saturation is below 92%." 32

Source

- GINA section: Section 9

- Page: 171, 172, 174, 180, 181, 183, 186
- Table/Figure: Box 9-2, Box 9-4, Box 9-6